

PRODUCT REQUIREMENTS DOCUMENT

AI Native Mastery Platform

Build credible skill mastery for learners regardless of connectivity

Document	Value
Status	Concept PRD for social-impact discovery, offline-first MVP planning, and Google Cloud prototype deployment
Version	0.10
Date	18 September 2026
Primary audience	Product, design, engineering, learning science, and pilot partners
Initial wedge	Data Problem-Solving and Automation

Working thesis

Build an offline-first, AI-native learning system that helps learners with limited connectivity develop practical skills, demonstrate what they can do independently, and continue learning without depending on continuous internet access.

Core promise

Make high-quality, evidence-based learning usable beyond well-connected classrooms. Help learners build, demonstrate, and retain practical capability without treating connectivity, device cost, or course completion as proxies for potential.

1 Executive Summary

Access to AI-enabled learning is expanding unevenly. Many products assume reliable broadband, capable devices, continuous cloud access, and prior familiarity with technical tools. Learners in rural and underserved communities may face unstable connectivity, expensive data, shared or low-memory devices, limited access to instructors, and fewer opportunities to demonstrate job-relevant skills. These constraints can exclude capable learners before learning begins and can widen existing education and opportunity gaps.

The proposed product is a Google Cloud-supported, offline-first mastery and evidence engine designed to reduce that access gap. Learners download a compact pathway when connectivity is available, then complete the core Diagnose, Learn, Practice, Verify, and Revisit loop on a personal device without an active connection. The system adapts to evidence and misconceptions, stores progress locally, distinguishes assisted learning from independent performance, and synchronizes through Firebase or Cloud Run when connectivity returns. Google Cloud extends the experience but is not required to complete the core loop after a module has been downloaded.

The MVP should validate two linked claims: first, that learners can complete a high-quality adaptive pathway on representative low-cost devices despite intermittent connectivity; second, that multi-modal evidence predicts independent performance better than course completion and conventional quiz scores. It should not attempt to become a general-purpose course catalog, credentialing body, employment evaluation system, or substitute for broader investments in teachers, devices, connectivity, and local education systems.

Generative AI is a substantive part of the learning and evidence experience. It acts as a misconception-aware tutor and qualitative evidence interpreter: it adapts explanations to the learner's error, creates bounded practice variations, asks follow-up questions about submitted work, and converts reasoning into reviewable evidence claims. It does not independently determine mastery. Deterministic validators, versioned rubrics, policy rules, and human escalation remain authoritative, and an authored fallback preserves the full learning loop when a suitable on-device model or connection is unavailable.

Recommended first product

- Target learner: motivated adults in rural or underserved communities who need practical data skills but may face intermittent connectivity, limited device capacity, uneven formal preparation, or limited access to instructors.
- Initial pathway: Data Problem-Solving and Automation, beginning with data literacy and spreadsheets, progressing through SQL, and introducing scripting when automation adds clear value.
- Core experience: downloadable diagnostic assessment, adaptive learning plan, a misconception-aware GenAI tutor in Learning Mode, bounded practice variation, dynamic follow-up questions in Proof Mode, hands-on tasks, delayed retention checks, and an evidence-backed skill profile. Every GenAI feature has an authored or deterministic fallback.
- Access model: Android-first personal-device experience that completes the core loop offline, uses compact downloadable or physically transferable content packs, and synchronizes opportunistically when connectivity returns.
- Pilot design: one domain and 100–300 learners, including a meaningful cohort using representative low-cost devices and intermittent connectivity, with an external transfer task scored blind to the product's mastery estimate.

Product decisions embedded in this PRD

Decision	Recommendation	Reason
Primary object	Skill and evidence, not course completion	Keeps the product aligned to actual capability.
Mastery output	Calibrated confidence with evidence history	Avoids false precision and makes claims auditable.
Modes	Separate Learning Mode and Proof Mode	Assistance is valuable for learning but weak evidence during verification.
Adaptation	Choose activities for learning value and information value	The next activity should teach or resolve an important uncertainty.
Initial participant	Learner pilot focused on access and learning value	Tests whether the experience is useful and reachable before adding institutional workflows.
Connectivity	Personal-device offline use first; local hub later	Validates the lowest-dependency access model before introducing hub operations.

Decision	Recommendation	Reason
Human role	Reviewer for disputed or high-stakes evidence	AI scoring alone should not determine consequential decisions.
Cloud role	Google Cloud-supported, not cloud-dependent	Firebase or Cloud Run provides deployment, synchronization, content delivery, and online AI enhancements while downloaded learning remains usable offline.

2 Problem Definition

2.1 Social and access problem

Learners do not receive equal access to high-quality adaptive learning. Rural location, unstable connectivity, expensive data, low-memory devices, limited instructor availability, and uneven prior preparation can restrict when and how a learner participates. Cloud-only AI experiences may reproduce these barriers by making continuous access a prerequisite rather than an enhancement.

2.2 Learning and evidence problem

Learners also cannot reliably tell which skills they understand, which they can apply without assistance, and which they are beginning to forget. Completion certificates and polished AI-assisted outputs may conceal specific misconceptions or weak independent performance, while static exams provide only a narrow snapshot.

2.3 System and program problem

- Education programs need credible, current evidence of capability without requiring invasive surveillance or permanent connectivity.
- Many digital learning experiences assume capable personal devices, reliable broadband, and continuous cloud services.
- Content and support are often centralized, making interruptions or high data costs disproportionately harmful to remote learners.
- Completion and time-on-task are treated as outcomes, while assessment often rewards recognition or memorization and rarely models prerequisite gaps, uncertainty, or forgetting.
- AI assistance is either banned indiscriminately or allowed without distinguishing supported production from independent capability; evidence remains fragmented across activities and time.

2.4 Social impact opportunity

Provide resilient learning infrastructure that continues working when the network does not. Give each learner a locally available pathway and a learner-controlled evidence record that answers: What can I do now? How strong is the evidence? What should I do next? When should this be checked again? The intended impact is greater access to future-ready skills and more credible demonstrations of capability for learners who are poorly served by cloud-dependent education.

2.5 Social impact boundary

The first pilot is English-only and cannot establish language inclusion or broad population impact. It must report who was reached, who was excluded, device and connectivity conditions, completion and transfer outcomes by access condition, and any burden shifted to learners or local facilitators. Offline delivery is one part of accessibility; it does not solve device affordability, disability access, local-language coverage, instructor shortages, or structural barriers on its own.

3 Product Vision and Principles

3.1 Vision

Learners should be able to develop and demonstrate practical capability regardless of whether they live near reliable broadband. Every learner has an adaptive tutor and an auditable skill profile that works through authentic practice and credible demonstration, while core learning and proof remain available without continuous internet access.

3.2 Product principles

Principle	Implication
Evidence before labels	Show why a skill is considered demonstrated or uncertain.
Learning and proof are different	Permit rich assistance in learning; constrain and record assistance in proof.
Uncertainty is useful	The system may say it lacks enough evidence and should seek the most informative next activity.
Transfer matters	Prefer novel application over repeated variants of memorized questions.
Learner agency	Let learners inspect, challenge, and supplement their evidence.
Low-surveillance integrity	Use task design, interaction evidence, and follow-up questioning before invasive monitoring.
Calibrated claims	Report confidence bands and evidence quality; avoid false precision.

Principle	Implication
Human accountability	No consequential decision about a person should rely only on an AI-generated mastery result.
Access under constraint	Design the core experience for intermittent connectivity and representative low-cost devices; treat cloud services as enhancement, not admission.

4 Users and Jobs to Be Done

User	Primary job	Initial priority
Learner	Help me learn and demonstrate practical skills even when connectivity, device capacity, or instructor access is limited.	Primary
Instructor or mentor	Show me where a learner is stuck and what evidence supports that conclusion.	Pilot support
Program owner	Determine whether the learning experience produces transferable capability.	Pilot partner
Community facilitator	Help learners provision content, recover from access problems, and continue learning without requiring permanent connectivity.	Later research
Content author	Map resources and tasks to skills and evidence types.	Internal first

Primary persona

A motivated adult learner in a rural or underserved community who wants to use data to solve practical problems, improve work, or enter a technical career. They may rely on intermittent

connectivity, a low-memory Android device, shared access to data or power, uneven formal preparation, and little or no prior programming experience.

Primary job story

When connectivity and learning support are limited, I want to keep developing practical data and automation skills, understand what I can do independently, and preserve credible progress so my location or access conditions do not determine my opportunity to learn.

4.1 Initial Pathway

The MVP will launch with Data Problem-Solving and Automation. The pathway teaches learners to frame a practical question, inspect and validate data, calculate or query an answer, communicate the result, and automate repeatable work. It is deliberately broader than a single tool and narrower than a data-science occupation.

Pathway progression

1. Data foundations: tables, data types, missing values, validation, privacy, and evidence-based conclusions.
2. Spreadsheet problem-solving: formulas, filters, summaries, charts, error checking, and reusable templates.
3. SQL investigation: selecting, filtering, joining, grouping, and validating results against source records.
4. Workflow automation: repeatable transformations, rule-based checks, and small scripts using an approved offline runtime.
5. Applied projects: locally relevant scenarios such as inventory, crop or clinic records, school attendance, household budgeting, and service operations, using synthetic or properly governed data.
6. Specialization bridges: learners may continue into software development, security operations, applied analytics, or AI-assisted work after demonstrating the relevant prerequisites.

MVP boundary

The pilot covers spreadsheet-like tables, CSV files, SQL, and limited Python or JavaScript automation. It does not claim to train a data scientist, software engineer, or cybersecurity professional. Cloud deployment, unrestricted packages, production databases, offensive security tooling, and large-scale machine learning remain outside the initial pathway.

Mastery evidence

Proof combines correct outputs with inspection of formulas or queries, validation checks, explanation of assumptions, recovery from planted errors, transfer to an unfamiliar dataset, and delayed reassessment. A polished chart or AI-generated script alone is not sufficient evidence of mastery.

Offline sandbox

The learner device provides a constrained workspace with a table viewer, spreadsheet-like editor, SQL runner backed by a local database, and an optional prepackaged scripting runtime. Starter datasets, tests, hints, and project history are stored in the content pack. Execution has time and memory limits, external network access is disabled by default, and work synchronizes only when a connection or local hub becomes available.

5 The Mastery Model

5.1 Skill graph

The core content structure is a directed graph of observable skills and prerequisites. A course is a curated path through that graph, not the canonical source of truth. Skills must be narrow enough to assess through behavior, such as “clean inconsistent records and justify each transformation,” rather than broad labels such as “knows data analysis.”

Skill field	Definition
Skill statement	An observable capability written as an action in a defined context.
Prerequisites	Skills whose absence materially reduces the chance of success.
Evidence requirements	Minimum mix of recall, explanation, application, transfer, and retention evidence.
Difficulty range	Expected complexity and scaffolding levels.
Misconceptions	Known wrong models the system should diagnose.
Assessment bank	Validated prompts, tasks, rubrics, and variants.
Resources	Explanations, examples, videos, documentation, and practice mapped to the skill.

5.2 Mastery state

The learner-facing state should remain understandable while the internal model may be probabilistic.

State	Meaning	Typical next action
Unknown	Not enough evidence to estimate current capability.	Short diagnostic
Exposed	The learner has encountered the idea but has not demonstrated it.	Guided example

State	Meaning	Typical next action
Developing	Some correct evidence exists, with important gaps or inconsistency.	Targeted practice
Demonstrated	The learner succeeds independently in representative contexts.	Novel transfer task
Mastered	Multiple strong, recent, independent evidence items support transfer.	Advance and schedule review
At risk	Earlier mastery evidence has aged or recent performance has declined.	Spaced retrieval check

5.3 Evidence model

Every evidence item records more than correctness. Its contribution depends on quality and context.

Dimension	Examples
Outcome	Correctness, rubric score, partial credit, error class
Independence	Unassisted, hints used, AI used, collaborator present
Cognitive demand	Recall, explain, apply, debug, design, transfer
Novelty	Seen pattern, variant, unfamiliar context
Recency	When the evidence was collected
Consistency	Agreement with other evidence across modalities
Authenticity	Interaction trace, follow-up explanation, artifact history
Rater confidence	Rubric reliability and model or human scoring confidence

A learner-facing mastery view should expose the evidence mix, most recent independent demonstration, known weaknesses, and next verification date. A numeric confidence may be shown only if calibrated and accompanied by plain-language meaning.

5.4 Next best activity policy

The selection engine balances five signals: expected learning gain, expected information gain, goal relevance, prerequisite leverage, and learner cost. It should also enforce variety and avoid repeatedly testing the same surface pattern.

Conceptual priority score: activity value = learning gain + information gain + goal relevance + prerequisite leverage – time and frustration cost. Initial MVP weights should be rules-based and observable; optimization can follow after sufficient outcome data exists.

6 Core Experience

6.1 Continuous loop

1. Diagnose: collect a compact baseline across prerequisites and target skills.
2. Learn: offer an explanation or resource suited to the learner's misconception and preference.
3. Practice: provide scaffolded activities with hints and feedback.
4. Verify: remove scaffolding, introduce a novel challenge, and collect independent evidence against an explicit rubric.
5. Revisit: schedule retrieval based on evidence strength, age, and observed decay.

The loop is not a fixed course sequence. It operates per skill. A learner may move directly from Diagnose to Verify when prior evidence is strong, return from Practice or Verify to Learn when a misconception appears, or revisit a previously mastered skill while continuing to develop other skills.

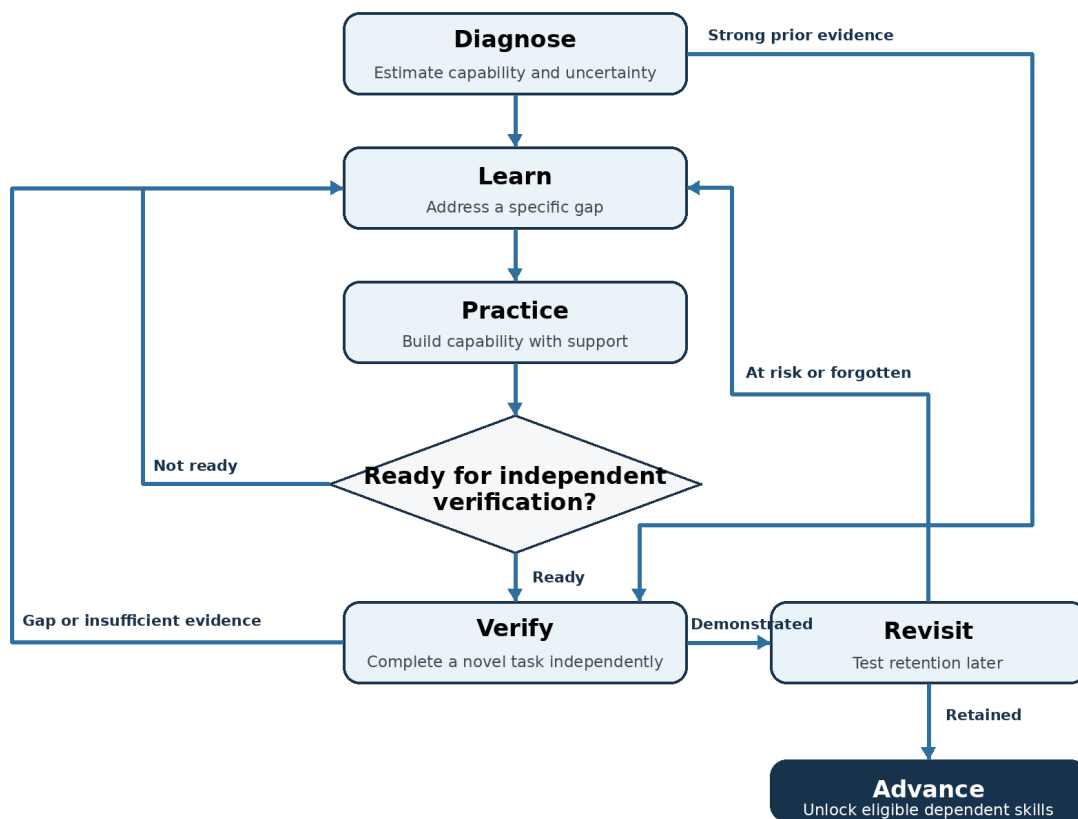


Figure 1 Mastery loop transitions

6.2 Learning Mode

Learning Mode is permissive and collaborative. Learners may ask questions, request alternate explanations, use examples, reveal hints, consult documentation, and work with the AI mentor. Activity events still record assistance so the system can distinguish learning progress from proof evidence.

6.3 Proof Mode

Proof Mode collects stronger evidence under declared constraints. It uses bounded sessions, novel tasks, randomized parameters, oral follow-ups, and artifact or interaction history. The interface tells the learner what assistance is allowed. A session may be paused or invalidated when the evidence cannot support the intended claim, but the learner should receive a clear reason and an appeal or retry path.

Proof method	What it tests	Integrity mechanism
Adaptive questions	Recall and conceptual discrimination	Item variation and follow-up on reasoning
Oral explanation	Mental model and independent reasoning	Dynamic probes tied to prior answers
Debugging task	Diagnosis and application	Novel defect and action trace
Hands-on build	Integrated performance	Milestones, version history, and defense questions
Transfer scenario	Generalization beyond practiced examples	Unfamiliar context with rubric-based scoring
Delayed review	Retention	Unannounced variant after an appropriate interval

6.4 Learner profile

- Goal and target role or outcome.
- Skill map with state, confidence language, and prerequisite relationships.
- Evidence timeline with modality, assistance level, rubric, date, and result.
- Misconceptions and skills requiring review.
- Recommended next activity with an explanation of why it matters.
- Learner controls to correct context, add evidence, request reassessment, or hide sharing.

6.5 Loop operation model

Every stage is a contract with an entry reason, a learner activity, evidence rules, and an exit decision. The next-best-activity policy selects a stage and activity for one or more skills; it does not simply advance the learner after completion.

Stage	Purpose	Learner experience	Evidence	Exit decision
Diagnose	Estimate current capability and uncertainty	Complete a short adaptive set of questions, explanations, or tasks	Correctness, reasoning, confidence, error pattern, difficulty, and assistance	Identify a gap, establish readiness for Verify, or remain Unknown
Learn	Introduce a capability or correct a specific misconception	Use a targeted explanation, example, dialogue, or demonstration	Self-explanation and formative checks; not strong proof evidence	Pass a comprehension check or receive a different explanation
Practice	Build capability with progressively reduced support	Complete varied exercises with feedback and optional hints	Accuracy, consistency, error recovery, difficulty, and assistance used	Meet a readiness rule, continue practice, or return to Learn
Verify	Test independent application and transfer	Complete a novel challenge under declared assistance constraints and explain the reasoning	Rubric score, independence, novelty, reasoning, provenance, and scorer confidence	Demonstrate the skill, collect more evidence, or return to a targeted gap
Revisit	Test retention after time has passed	Complete a delayed retrieval or transfer task with meaningful variation	Retention interval, independence, transfer, and contradictory evidence	Keep current, mark At risk, or reopen Learn or Practice

6.6 Evidence processing after each activity

1. Record the result with skill, difficulty, novelty, permitted and actual assistance, time, modality, rubric, content version, and provenance.
2. Validate the evidence. Reject, quarantine, or request review when the activity is incomplete, corrupted, outside its permitted conditions, or scored with low confidence.
3. Update the mastery estimate. Independent novel evidence carries more weight than assisted practice; recent contradictory evidence can lower confidence.
4. Locate the remaining uncertainty. Distinguish whether the unresolved question concerns recall, explanation, application, transfer, retention, or evidence authenticity.
5. Select the next activity by expected learning gain, information gain, goal relevance, prerequisite leverage, learner cost, offline availability, and required modality.

The learner should see a concise reason for the selection. Example: You can summarize a clean table, but there is not enough evidence that you can detect and repair inconsistent records independently. Your next activity is an unassisted data-cleaning task.

6.7 Initial transition rules

The MVP uses transparent, versioned rules. A language model may help interpret an answer, but it does not control progression without the policy layer.

Current condition	Next stage	Reason
Diagnostic reveals a misconception or missing prerequisite	Learn	Target the specific explanatory gap before more testing
Diagnostic provides strong recent independent evidence	Verify	Confirm transfer without unnecessary instruction
Comprehension check succeeds	Practice	Move from conceptual understanding to supported performance
Practice remains inconsistent or hint-dependent	Practice or Learn	Increase variation or repair the recurring misconception
Practice succeeds across sufficiently difficult variants with little assistance	Verify	Test whether performance transfers without scaffolding
Verify meets the rubric with valid independent evidence	Revisit	Schedule retention while unlocking eligible dependent skills
Verify fails in a diagnosable way	Learn or Practice	Remediate the demonstrated gap instead of repeating the same test
Delayed evidence is weak, old, or contradictory	Revisit or Learn	Refresh the skill and reduce overconfidence

Thresholds are configuration, not universal truths. The team must calibrate readiness and mastery thresholds against blind-scored external transfer tasks and revise them by domain when evidence supports the change.

6.8 End to end example

Target skill: detect and repair inconsistent values in a small operational dataset.

Stage	What happens	System decision
Diagnose	The learner reviews an inventory table, notices duplicate item names, but treats inconsistent units as equivalent.	Infer a misconception about data types and unit normalization, then select a targeted explanation.

Stage	What happens	System decision
Learn	The learner studies a worked comparison of kilograms and grams, then explains why values must be normalized before aggregation.	A short comprehension check indicates that the misconception may be corrected.
Practice	The learner cleans three datasets with decreasing hints, validates totals, and recovers from a planted formatting error.	The readiness rule is met because varied datasets are corrected consistently with little assistance.
Verify	The learner independently cleans an unfamiliar stock-count dataset, writes a summary query, and explains each transformation.	The rubric supports Demonstrated because correctness, validation, independence, explanation, and transfer requirements are met.
Revisit	Fourteen days later, the learner receives a clinic-supplies dataset with different unit and duplicate-record problems.	Success keeps the skill current; partial success marks it At risk; failure reopens the relevant gap.

6.9 Offline execution contract

- Diagnose uses downloaded items, rubrics, and deterministic adaptation rules.
- Learn always has an authored text-first resource; local AI is optional and may provide explanations or dialogue on capable hardware.
- Practice selection and mastery updates run locally and never depend on a network response.
- Verify uses locally available secure tasks, records permitted assistance, and appends evidence before showing completion.
- Revisit uses the local schedule and device notifications. The learner can complete it while still offline.
- Complex AI scoring may remain provisional until local-hub or cloud review, but progress and evidence are never discarded. Synchronization preserves the original event and policy versions.

6.10 Generative AI assisted learner journey

Generative AI changes the response to the learner rather than replacing the mastery engine. Each invocation receives a bounded skill definition, approved source excerpts, the relevant learner attempt, known misconceptions, permitted assistance, and an output constraint. The learner can inspect the resulting explanation or question, while the system records the model, prompt policy, source bundle, inference location, and assistance level.

Moment	Generative AI contribution	Authoritative control
Diagnose	Interpret explanations or work traces and propose likely misconceptions.	Validated items and deterministic checks establish correctness; model labels remain hypotheses.
Learn	Create a targeted explanation, analogy, example, or short Socratic exchange grounded in approved material.	The authored lesson is always available; the model may not introduce unapproved learning objectives.
Practice	Generate bounded variants and progressive hints based on the current error pattern.	Templates, validators, difficulty limits, and assistance logging constrain the activity.
Verify	Ask a dynamic defense question tied to the learner's artifact or reasoning.	Novel task selection, allowed assistance, rubric scoring, and mastery transitions remain versioned and reviewable.
Evidence review	Transform qualitative responses into structured, traceable evidence claims with uncertainty.	Rules combine evidence; low-confidence or consequential cases require human review.

Offline hierarchy. Low-cost or unsupported devices use authored explanations, pre-generated variants, and deterministic feedback. Supported devices may use short, single-turn on-device generation. Connected sessions may use richer Gemini dialogue, multimodal review, or speech. Loss of either the model or the network must degrade the experience, not block it.

7 Primary User Flows

7.1 New learner

1. Select a concrete goal and initial module.
2. Review what data will be collected and how evidence may be used.
3. Complete a 15–25 minute adaptive diagnostic.
4. Receive a skill map that distinguishes observed evidence from unknown areas.
5. Start the recommended learning activity and see why it was chosen.
6. Complete an independent challenge after enough guided practice.
7. Review the evidence-backed result and scheduled follow-up.

7.2 Daily return

The learner sees one recommended action, its expected duration, whether it is for learning or proof, and the reason it is prioritized. They may choose an alternative while the system records preference and adapts within goal constraints.

7.3 Instructor review

An instructor inspects cohort-level gaps and individual evidence, filters low-confidence or disputed assessments, and can annotate or override a result with a reason. Overrides remain visible in the audit history.

7.4 Prototype demonstration flow

The working prototype must demonstrate that Google Cloud integration and offline accessibility reinforce rather than contradict each other. The evaluator flow is:

1. Connect and download a compact Data Problem-Solving and Automation module.
2. Disconnect the device from the internet.
3. Complete representative Diagnose, Learn, Practice, Verify, and Revisit activities while progress and evidence are committed locally.
4. Reconnect and synchronize progress through Firebase or a Cloud Run service.
5. Show at least one meaningful Gemini or Gemma capability, such as targeted explanation, evidence review, or content-pack preparation.
6. Confirm that no completed learning evidence is lost across interruption and synchronization.

8 Offline First and Rural Deployment

8.1 Product requirement

A learner must be able to complete the core Diagnose, Learn, Practice, Verify, and Revisit loop without an active internet connection. Important actions are committed locally before any network request. Connectivity enhances the experience but must not be required to open downloaded material, save progress, complete eligible assessments, update the local mastery state, or schedule review.

8.2 Deployment model

Layer	Offline responsibility	Connected responsibility
Learner device	Content, activities, local profile, evidence log, mastery updates, sync queue, and optional small AI	Download updates, back up events, and receive reviewed results
School or community hub	Local Wi-Fi access, shared content, teacher dashboard, device sync, backups, and optional shared AI	Exchange compact updates with central services when a connection is available
Cloud services	Not required for an active offline session	Content publishing, cross-site synchronization, stronger AI, human review, analytics, and model updates
Physical media	Distribute signed content, model, and software bundles by USB, SD card, or portable drive	Prepared from an authorized connected environment

8.3 Hardware tiers

Tier	Expected experience	Product behavior
Low-cost phone	Text, compressed images, quizzes, deterministic adaptation, progress, and selected audio	AI is optional; fall back to authored explanations and rule-based scoring
Capable phone or tablet	Core experience plus a small local language model for hints and constrained dialogue	Display model download size and permit removal without losing learning data
Local learning hub	Shared content library, classroom dashboard, local sync, and more capable AI services	Continue serving devices over local Wi-Fi when the internet is unavailable
Connected environment	Full synchronization, difficult scoring, human review, and large updates	Never block offline work while cloud processing is pending

8.4 Synchronization and distribution

- Use an append-only evidence log with stable event identifiers so retries do not duplicate progress.
- Prioritize small evidence and profile updates before content, video, or model downloads.
- Resume interrupted transfers and show pending, synchronized, failed, and conflict states in plain language.
- Version skills, activities, rubrics, policies, content packs, and AI models; preserve the version used for historical evidence.
- Support internet, local-network peer, school-hub, USB, SD-card, and portable-drive update paths.
- Resolve ordinary progress merges automatically; route contradictory profile or scoring changes for review.

8.5 Offline Proof Mode limitations

Offline Proof Mode can produce credible evidence through novel tasks, randomized parameters, interaction history, oral defense, delayed reassessment, and a device-signed event log. It cannot guarantee identity or the absence of outside help. Learner-facing claims must describe the recorded conditions. High-impact evidence may remain provisional until it is synchronized, validated, or reviewed.

8.6 Rural accessibility requirements

- Android-first support with testing on representative older and low-memory devices.
- Text-only and low-data content packs, transcripts, compressed media, and visible download sizes.
- No progress loss during power interruption, application termination, or unstable connectivity.
- Battery-aware synchronization and configurable Wi-Fi-only downloads.

- Local-language content packs and equivalent non-voice paths where speech models are unavailable or unreliable.
- Administrators can provision multiple devices locally without requiring individual high-bandwidth downloads.

8.7 Google Cloud supported offline architecture

The product uses a local-first execution boundary. Cloud services are meaningful parts of the prototype, but temporary loss of connectivity must not block the downloaded core learning loop.

- Learner device: stores downloaded content packs, local mastery state, activity events, pending evidence, and scheduled reviews. Eligible activities and deterministic adaptation rules run without a network connection.
- Firebase: may host the web or PWA shell, manage authentication, distribute versioned content packs, and synchronize learner-controlled records when a connection returns.
- Cloud Run: may provide heavier Gemini services, content-pack preparation, complex or provisional evidence review, analytics, and conflict-resolution support.
- Gemma or equivalent on-device model: may provide bounded offline explanations or coaching when feasible on the supported device tier. A deterministic non-generative fallback must remain available when local inference is unsupported.
- Synchronization boundary: connected services receive versioned events rather than replacing the local record. The client shows pending, synchronized, failed, and conflict states and never silently discards accepted offline work.

9 MVP Scope

9.1 In scope

Capability	MVP requirement
Skill graph	Data Problem-Solving and Automation with 30–60 observable skills spanning data foundations, spreadsheets, SQL, and limited scripting.
Diagnostic	Adaptive baseline using a bounded item and task bank.
AI mentor	Dialogue grounded in approved resources, skills, and misconceptions.
Adaptive practice	Rules-based activity selection with transparent reasons.
Proof Mode	Adaptive quiz, oral explanation, and one hands-on task type.
Mastery estimate	Evidence-weighted state with uncertainty and decay.
Evidence record	Timestamped events with rubric, assistance, provenance, and scorer confidence.
Dashboard	Learner skill map; basic instructor review queue and cohort summary.
Retention	At least one delayed reassessment schedule.

Capability	MVP requirement
Evaluation	External transfer task and calibration analysis.
Offline operation	Android-first local data, downloadable content pack, sync queue, and resilient restart.
Future hub readiness	Content, evidence, and synchronization formats remain compatible with a later local hub, but hub hardware and operations are not required for the first deployment.
Offline sandbox	Constrained table, SQL, and scripting workspace with starter datasets, deterministic tests, local project history, resource limits, and no external network access by default.
Google Cloud prototype deployment	Deploy a functional client through Firebase or Cloud Run, use Firebase or Cloud Run for real synchronization or online processing, and integrate a meaningful Gemini or Gemma capability. The downloaded core loop must remain usable when the network is disabled.
Generative AI tutor and evidence interpreter	Use Gemini or Gemma to generate grounded targeted explanations, bounded hints or variants, dynamic follow-up questions, and reviewable evidence interpretations. Preserve authored fallbacks and keep mastery decisions outside the generative model.

9.2 Explicitly out of scope

- AI-generated video production as a core capability.
- A broad catalog of unrelated courses or mentors.
- Accredited certification or claims equivalent to a degree or professional license.
- Employment screening, worker evaluation, or other consequential decision integrations.
- Remote proctoring based on face, gaze, emotion, or biometric inference.
- Open-ended assessment of every professional domain.
- Fully autonomous generation and release of high-stakes assessment items.
- Public skill profiles by default.
- Requiring an on-device generative model for the core learning loop.
- High-quality AI video generation on learner devices.

9.3 MVP assumptions

- The first domain can be decomposed into observable skills with usable rubrics.
- A short interaction history plus dynamic follow-ups can provide useful integrity signals without invasive surveillance.
- Learners accept separate assisted and independent modes when the benefit is clearly explained.
- Pilot partners can provide or approve source material, tasks, and an external outcome measure.
- The pilot can obtain representative low-cost Android devices and at least one local-hub configuration for field testing.

10 Functional Requirements

ID	Requirement	Priority
FR1	Create and version skills, prerequisites, misconceptions, rubrics, resources, and evidence requirements.	Must
FR2	Run a diagnostic that adapts difficulty and skill coverage while respecting a time budget.	Must
FR3	Recommend one next activity and display a learner-readable reason.	Must
FR4	Support Learning Mode with grounded explanations, hints, examples, and feedback.	Must
FR5	Support Proof Mode with declared assistance constraints and evidence capture.	Must
FR6	Score structured and open responses against versioned rubrics, with confidence and review thresholds.	Must
FR7	Update mastery state only from valid evidence and preserve the full event history.	Must
FR8	Schedule reassessment based on elapsed time, prior strength, and recent contradictory evidence.	Must
FR9	Allow learners to inspect evidence, request a retry, and dispute a scored result.	Must
FR10	Provide reviewer queues for low-confidence, disputed, or high-impact evidence.	Should
FR11	Export a learner-controlled evidence summary with scope and limitations.	Should
FR12	Support human-authored and approved content before allowing generated assessment content.	Must
FR13	Complete the core learning and mastery loop without an active internet connection.	Must
FR14	Persist local events before network submission and synchronize them idempotently when connectivity returns.	Must
FR15	Download, verify, update, and remove versioned content packs without deleting learner evidence.	Must

ID	Requirement	Priority
FR16	Distribute approved updates through cloud, local hub, peer network, or physical media.	Should
FR17	Provide deterministic non-AI fallbacks for explanations, recommendations, and eligible scoring.	Must
FR18	Show offline availability, storage cost, update status, and pending synchronization clearly.	Must
FR19	Apply versioned stage-transition rules and record the reason for every next-stage and next-activity decision.	Must
FR20	Run approved spreadsheet-like, SQL, and scripting activities offline in a constrained sandbox and record commands, revisions, tests, assistance, and outputs as evidence.	Must
FR21	Generate a targeted explanation from an approved source bundle, detected misconception, learner attempt, and declared reading level.	Must
FR22	Generate bounded hints or practice variants without revealing protected answers or changing the skill objective.	Should
FR23	Generate a follow-up question tied to the learner's submitted artifact or explanation and record that AI assistance occurred.	Must
FR24	Return traceable evidence interpretations with model, policy, source bundle, inference location, uncertainty, and review status.	Must
FR25	Select authored, on-device, or cloud GenAI behavior from explicit capability and connectivity checks, with no hidden dependency on cloud inference.	Must

11 Nonfunctional Requirements

Area	Requirement
Explainability	Every mastery update and recommendation must be traceable to evidence and a policy version.
Latency	Conversational feedback should begin within 3 seconds; complex scoring may complete asynchronously with status.

Area	Requirement
Reliability	No evidence event may be silently lost; scoring retries must be idempotent.
Accessibility	Target WCAG 2.2 AA, keyboard navigation, captions or transcripts, and equivalent non-voice alternatives.
Privacy	Private by default, purpose-limited collection, configurable retention, deletion, and export.
Security	Encryption in transit and at rest, tenant isolation, role-based access, and audit logs.
Model governance	Version prompts, models, rubrics, and policies; support rollback and retrospective evaluation.
Localization	Architecture supports future localized content and rubrics; the first pilot is English only.
Offline resilience	No completed response or accepted evidence may be lost after power loss, process termination, or interrupted synchronization.
Data efficiency	Prioritize compact event sync; make large downloads explicit, resumable, and optionally Wi-Fi-only.
Device support	Set and test minimum Android, memory, storage, and performance targets using representative rural-deployment hardware.
Cloud deployment	The prototype is deployed on Google Cloud through Firebase or Cloud Run. Cloud unavailability must not prevent opening downloaded content, saving eligible work, updating local mastery state, or scheduling review.

12 AI and Data Design

12.1 AI responsibilities

Component	AI role	Required guardrail
Tutor	Generate grounded, misconception-specific explanations, questions, examples, and feedback from bounded learner context.	Approved source bundle, prompt version, assistance logging, short-output limit, and authored offline fallback.
Activity selector	Rank eligible next activities.	Use observable features and log the reason; begin with rules.
Assessment generator	Draft controlled variants from approved templates; never release high-stakes items autonomously.	Human-approved blueprint, leakage checks, and item review sampling.
Scorer	Interpret open responses and oral transcripts into rubric-linked evidence claims with uncertainty.	Confidence threshold, second-pass checks, and human escalation.
Mastery updater	Combine evidence into a current estimate.	Run locally through a calibrated, versioned model with monotonic and policy constraints.
Integrity assistant	Identify inconsistencies and trigger follow-up questions.	Never infer deception from identity, emotion, gaze, or accent.

12.2 Generative AI operating model

The primary learner-facing GenAI capability is misconception-aware tutoring. The primary evidence capability is dynamic probing and qualitative evidence interpretation. These uses are substantive because the generated response depends on the learner's actual reasoning and changes the next interaction. Generic chat, decorative summaries, or a disconnected chatbot do not satisfy this requirement.

Capability	Bounded input	Expected output	Decision authority
Targeted tutor	Skill, misconception, attempt, approved excerpts, reading level	Explanation, analogy, worked example, comprehension question	Advisory; authored lesson remains available
Practice assistant	Skill template, constraints, recent errors, hint budget	Variant, hint, feedback, or counterexample	Validators determine correctness and readiness
Proof probe	Submitted artifact, action trace, rubric dimension	One focused follow-up question	Rubric and review policy determine evidence weight
Evidence interpreter	Response, artifact summary, provenance, rubric	Claims, supporting observations, uncertainty, escalation flag	Mastery updater accepts only policy-valid evidence
Authoring assistant	Approved curriculum and assessment blueprint	Draft explanation, item variant, rubric language, or audio script	Human approval required before release

Generation controls. Prompts and source bundles are versioned. Cloud responses should use schema-constrained output where supported. On-device Android generation is treated as optional and capability-gated because current Firebase AI hybrid support is experimental, device-limited, single-turn, and does not support structured output. A parser may validate simple local responses, but failure must route to an authored fallback rather than silently invoking cloud inference while the learner believes the session is offline.

12.3 Core entities

Entity	Selected fields
Skill	id, statement, domain, prerequisites, level, version
Activity	skills, modality, difficulty, duration, mode, allowed assistance
Evidence event	learner, skill, outcome, rubric, independence, novelty, timestamp, provenance
Mastery state	learner, skill, state, probability range, updated time, policy version
Misconception	skill, diagnostic signals, remediation links
Recommendation	candidate set, selected activity, reason, policy version
Stage decision	current stage, trigger evidence, eligible transitions, selected transition, reason, policy version
Review	trigger, reviewer, decision, rationale, prior and revised score
Sync event	event id, device, local sequence, payload version, status, attempts, acknowledgment
Content bundle	bundle id, locale, skills, version, size, manifest, signature, dependencies

12.4 Initial mastery implementation

Start with an interpretable evidence-weighting model rather than an opaque end-to-end predictor. Correct independent transfer evidence should carry more weight than assisted practice. Contradictory recent evidence should reduce confidence. Older evidence should decay at a skill-specific rate. Prerequisites may influence recommendations, but a weak prerequisite should not automatically overwrite direct evidence for a downstream skill.

Once the pilot produces enough longitudinal data, compare Bayesian knowledge tracing, item-response approaches, and learned sequence models. Promotion should require better calibration and prediction of external transfer—not merely higher fit to internal quiz outcomes.

12.5 Content and assessment quality

- Maintain a human-approved assessment blueprint specifying skill coverage and cognitive demand.
- Version rubrics and preserve the rubric used for every historical score.
- Measure item difficulty, discrimination, exposure, ambiguity, and differential performance.
- Keep a secure holdout bank for transfer evaluation and refresh compromised items.
- Do not train or tune on the same responses used for final claims without a documented split.

12.6 Google AI deployment boundary

The prototype must use Gemini, Gemma, or an eligible Google agentic platform for a substantive product capability. Integration is substantive when the model changes an explanation, review, recommendation, or content artifact that a learner or reviewer can inspect. A logo, unused endpoint, or cloud-hosted static page does not meet this intent.

Cloud AI outputs that affect mastery evidence remain versioned, reviewable, and distinguishable from deterministic local decisions. When a cloud-only review is unavailable, the device preserves the evidence as pending or provisional and allows the learner to continue with eligible offline work.

13 Trust Integrity and Responsible Use

13.1 Integrity posture

The product should acknowledge that perfect authorship detection is not possible. It should make bounded claims about the conditions under which evidence was collected. Strong proof comes from converging behavior: solving, explaining, responding to novel probes, and retaining the skill—not from a single detector score.

13.2 Learner rights

- Know whether an activity is for learning, practice, or proof.
- Know what assistance is allowed and what data is collected.
- See the evidence behind a mastery claim and the limitations of that claim.
- Challenge an automated score and obtain human review where consequences are meaningful.
- Control sharing and revoke access to a profile where contractually and legally possible.
- Use an equivalent alternative when voice or a specific modality creates an accessibility barrier.

13.3 Product boundary

The product is a learner-facing education and mastery system. It will not provide applicant screening, hiring recommendations, worker evaluation, promotion decisions, or automated eligibility decisions. Mastery evidence is designed to guide learning and reflection, not to rank people for consequential external decisions.

14 Success Metrics and Validation

14.1 North star

Verified durable mastery: the proportion of target skills that a learner demonstrates independently on a novel task and retains at a delayed check. This is more meaningful than lessons completed or time spent.

14.2 Metric hierarchy

Category	Metric	MVP target or decision rule
Validity	Correlation or predictive performance against blind-scored external transfer tasks	Materially outperform completion and conventional quiz-score baselines
Calibration	Observed success by predicted mastery band; Brier score or expected calibration error	No material overconfidence; predefine acceptable calibration bands
Learning	Pre-to-post gain on secure equivalent forms	Positive gain versus comparison condition
Retention	Delayed independent performance after 2–4 weeks	Better retention than baseline experience
Efficiency	Verified mastery gained per learner hour	Improve without lowering external performance
Coverage	Share of mastery claims supported by more than one evidence modality	At least 70% for claims shown as Mastered
Trust	Learner-reported clarity and fairness; dispute and overturn rates	Monitor by group and modality
Engagement	Diagnostic completion, weekly active learning, return for spaced review	Secondary; never substitute for validity
Offline reliability	Sessions completed without internet, recovery after interruption, and unsynchronized-event loss	No accepted evidence loss; predefined crash-recovery and sync-success gates
Access efficiency	Data transferred, storage used, battery impact, and time to provision a learner	Meet targets on representative low-cost hardware and intermittent networks
Reach	Eligible learners who can install, provision, and begin the pathway under target	Predefine a successful-access threshold and report failure reasons by device and access

Category	Metric	MVP target or decision rule
	device and connectivity conditions	condition
Access parity	Difference in completion, verified mastery, and retention between intermittent-connectivity and reliably connected learners	No material disadvantage after accounting for baseline skill and declared learner context
Learner burden	Data cost, storage, battery use, travel or facilitator dependence, and time lost to synchronization	Stay within predefined field-tested limits and do not shift hidden operating costs to learners

14.3 Pilot hypotheses

Hypothesis	Test	Pass signal
The mastery model predicts transfer better than completion.	Compare external-task prediction using mastery estimate, quiz score, and completion.	Pre-registered improvement with uncertainty reported.
Adaptive selection improves efficiency.	Randomize adaptive path versus fixed sequence.	Equal or better transfer performance in less time.
Multi-modal evidence improves calibration.	Ablate oral, hands-on, and delayed evidence.	Full model reduces calibration error on holdout learners.
Proof Mode distinguishes assisted output from independent capability.	Compare Learning Mode artifacts with novel Proof Mode tasks.	Meaningful within-learner separation where assistance was substantial.
Learners accept the distinction between modes.	Interview and survey after use.	Most learners understand the purpose and view constraints as proportionate.
Offline access preserves outcomes.	Compare offline-first and connected cohorts on equivalent content and proof tasks.	No material loss in transfer performance after accounting for learner context.

Hypothesis	Test	Pass signal
Personal-device offline delivery broadens successful access.	Measure installation, provisioning, completion, learner burden, and failure modes under representative low-cost-device and intermittent-connectivity conditions.	Target learners can complete the core loop without a hub and without unacceptable data, storage, battery, or support burden.

14.4 Guardrail metrics

- False-confidence rate: learners labeled Mastered who fail the external task.
- Group calibration gaps and accessibility-related completion gaps.
- Automated-score disagreement with trained human raters.
- Dispute rate, review latency, and score-overturn rate.
- Hallucinated or unsupported tutor responses per audited session.
- Assessment leakage, repeated item exposure, and compromised-task rate.
- Learner stress, perceived surveillance, and withdrawal attributed to Proof Mode.
- Lost or duplicated evidence events, unresolved sync conflicts, and days of unsynchronized data.
- Performance, battery, storage, and failure rates by supported device tier.

15 Analytics and Experimentation

Events should support reconstruction of the learner journey without capturing unnecessary sensitive data. Minimum events include activity recommended, activity started, assistance requested, response submitted, rubric scored, evidence accepted or rejected, mastery updated, recommendation overridden, review requested, review decided, and retention check completed.

Experiments must separate learning outcomes from engagement. A feature that increases session length but reduces transfer or increases overconfidence is a failure. Analysis plans should be written before viewing results for core validity claims.

16 Delivery Plan

Phase	Duration	Deliverable	Exit criterion
0 Discovery	3–4 weeks	Pathway decomposition, social-impact problem validation, learner research, sandbox feasibility, access constraints, and pilot dataset plan	Bounded module, measurable external outcome, intended learner profile, and target access conditions approved
1 Instrumented prototype	6–8 weeks	Offline diagnostic, mentor fallback, evidence schema, local persistence, manual review, and skill map	End-to-end offline sessions run with 15–30 learners on representative devices and connectivity conditions
2 Pilot MVP	8–12 weeks	Proof Mode, mastery updater, content packs, personal-device synchronization, retention, and instructor review	100–300 learner pilot completed with meaningful representation of the intended rural or underserved cohort
3 Validation	4–6 weeks	Calibration, transfer, subgroup, and qualitative findings	Go, revise, or stop decision against pre-registered gates
4 Productization	TBD	Authoring tools, tenant controls, stronger review operations	Only after validity and demand are supported

17 Risks and Mitigations

Risk	Why it matters	Mitigation
False mastery confidence	A polished score could overstate real capability.	Calibrate to external tasks, show uncertainty, require converging evidence.

Risk	Why it matters	Mitigation
AI scoring inconsistency	Equivalent answers may receive different judgments.	Versioned rubrics, repeated scoring checks, human review thresholds.
Assessment leakage	Generated or reused tasks may become searchable or memorized.	Secure holdouts, variants, exposure tracking, dynamic defense questions.
Goodhart's law	Learners optimize the score rather than the capability.	Use varied novel tasks and keep outcome evaluation separate.
Bias and accessibility	Voice, language, disability, or culture may affect performance unrelated to skill.	Alternate modalities, subgroup calibration, accommodations, appeal.
Surveillance creep	Integrity features can become invasive and damage trust.	Prohibit biometric or emotion inference; minimize collection and state limits.
Content quality	Incorrect explanations or ambiguous questions teach the wrong thing.	Approved sources, authoring workflow, audit sampling, issue reporting.
Cold start	The model lacks enough evidence for a new learner or skill.	Explicit Unknown state, compact diagnostic, conservative claims.
Out-of-scope consequential use	Learning evidence could be repurposed to rank or evaluate people outside the intended educational context.	Prohibit consequential decision integrations, minimize external claims, preserve learner control, and enforce purpose limitations.
Cost and latency	Voice, long context, and repeated scoring can be expensive.	Tiered models, caching, structured rubrics, asynchronous review.
Low-end device limits	Local AI, media, or large content packs may exceed memory, storage, or battery capacity.	Hardware tiers, small bundles, authored fallbacks, hub inference, and field benchmarks.
Sync failure	Long offline periods or retries may lose, duplicate, or conflict with evidence.	Append-only event log, stable IDs, acknowledgments, resumable transfer, and conflict review.
Unequal experience	Connected learners may receive stronger AI or faster review than offline learners.	Measure outcome parity, keep core pathways equivalent, and disclose pending enhanced review.
Hub operations	Local servers can fail or become difficult for schools to maintain.	Simple appliance design, health checks, replaceable storage, administrator training, and offline recovery media.

Risk	Why it matters	Mitigation
Cloud dependency weakens the offline promise	A prototype can appear offline-first while essential AI, saving, or verification still fails without a connection.	Test the complete downloaded core loop in airplane mode, treat cloud review as pending rather than blocking, and demonstrate recovery and synchronization after reconnecting.

18 Product Decision Register

Decided items are approved product constraints. Recommended items remain proposals until confirmed through discovery, feasibility testing, or pilot governance review.

Decision	Options	Direction
1 First pilot module	A. Inventory and small-business operations; B. Community-service records; C. Personal finance and budgeting.	Recommended: A. It supports realistic spreadsheet, SQL, validation, and automation tasks using safe synthetic data, with broad relevance and clear transfer measures.
2 Initial participant	Learner; instructor-led cohort; community learning program; university learning program.	Decided: Learner. Begin with direct learner access and learning value; treat education institutions only as possible later delivery or support partners.
3 Mastery threshold	A. One universal score; B. Fully domain-specific thresholds; C. Shared evidence minimum plus domain-specific rubric thresholds.	Recommended: C. Require common evidence qualities such as independence, novelty, and retention, then calibrate task thresholds for each skill domain.
4 Proof Mode friction	A. Minimal constraints; B. Moderate bounded sessions and dynamic follow-ups; C. High-control remote proctoring.	Recommended: B. Use novel tasks, declared assistance rules, interaction history, and short defense questions; avoid biometric or surveillance-heavy controls.
5 Retention interval	A. 7 days; B. 14 days; C. 28 days; D. Multiple intervals.	Recommended: D, with day 14 as the primary MVP outcome and day 28 as a smaller durability check where pilot duration permits.
6 Evidence portability	A. Product-only report; B. Learner-controlled signed export; C. Immediate adoption of an external credential standard; D. Defer portability.	Recommended: B. Export a human-readable summary plus signed structured evidence; map to external standards only after the evidence model stabilizes.
7 Mandatory human review	A. Review every open response; B. Review only disputes; C. Review low-confidence, disputed, high-impact, and sampled cases.	Recommended: C. Target review within three business days during the pilot and audit a random sample of otherwise accepted automated decisions.
8 Data retention	A. Indefinite retention; B. Fixed short retention; C. Active-account retention plus deletion after inactivity; D. Learner-selected retention only.	Recommended: C. Retain while active, default to deletion 24 months after inactivity, and provide export and deletion controls subject to documented legal obligations.
9 Minimum device profile	A. Android 8 with 2 GB RAM; B. Android 10 with 3 GB RAM and 2 GB free storage; C. Android 12 with 4 GB RAM.	Recommended: B as the supported baseline, while testing a reduced text-and-SQL experience on selected 2 GB devices before making a broader claim.
10 First deployment model	Personal-device offline use; school or community hubs; both from launch.	Decided: Personal-device offline use. Design formats for future hubs, but do not make hub hardware or administration an MVP dependency.
11 Pilot language	English only; English plus Filipino; English plus multiple local languages.	Decided: English only for the pilot. Preserve localization architecture and equivalent non-voice paths for later expansion.
12 Provisional offline evidence	A. No time limit; B. Fixed expiry; C. Local reviewer finalization; D. Tiered rule based on intended use.	Recommended: D. Never discard learning progress, but prevent externally shared verified claims after 30 unsynchronized days unless a qualified local reviewer finalizes them.
13 Google Cloud integration	A. Firebase only; B. Cloud Run only; C. Hybrid Firebase and Cloud Run.	Recommended: A for the prototype. Use Firebase for hosting, authentication, content distribution, offline-aware synchronization, and the minimum backend needed for the Gemini or Gemma capability. Add Cloud Run only when heavier Gemini processing or independent scaling is demonstrably necessary.

Decision	Options	Direction
14 GenAI product role	A. Generic chatbot; B. Automated mastery judge; C. Misconception-aware tutor and evidence interpreter with deterministic authority	Recommended: C. It makes GenAI essential to personalization and qualitative evidence while preserving offline continuity, auditability, and learner rights.

19 Launch Decision Gates

Proceed beyond pilot only if the product demonstrates all of the following:

- The mastery estimate predicts blind-scored transfer meaningfully better than completion and ordinary quiz scores.
- Calibration is acceptable overall and no critical subgroup shows unexplained severe overconfidence.
- Learners understand the difference between Learning Mode and Proof Mode and do not experience the integrity design as disproportionate surveillance.
- AI tutor and scorer errors remain within predefined operational thresholds, with effective review and correction.
- Target learners report that the pathway is useful, understandable, and worth continuing, and participation is not dependent on an institutional deployment.
- The full core loop functions on supported low-cost devices without internet and survives power, process, and synchronization interruptions without accepted evidence loss.
- Offline learners achieve comparable transfer outcomes and are not systematically disadvantaged by delayed AI or human review.
- The pilot includes meaningful representation of the intended rural or underserved learner profile and reports exclusions and recruitment bias.
- Installation, provisioning, offline completion, synchronization, data use, storage, battery impact, and support burden meet predefined field-tested limits on representative low-cost devices.
- Intermittent-connectivity learners show no material disadvantage in verified mastery or retention after accounting for baseline skill and declared context.

If predictive validity fails, the team should not compensate by adding more content modalities. It should revise the skill definitions, evidence design, or measurement model and repeat validation.

Appendix A Example Skill Module

Illustrative domain: Data Problem-Solving and Automation. This example decomposes a broad pathway into observable skills that can be practiced and verified offline.

Skill	Evidence sequence	Mastery requirement
Validate a dataset	Inspect schema and quality → identify defects → justify corrections	Independent detection and repair across two unfamiliar datasets

Skill	Evidence sequence	Mastery requirement
Query related tables	Trace keys → write a join → diagnose duplicate or missing rows	Independent query and validation on a novel schema
Automate a repeatable workflow	Describe steps → implement rules or script → test edge cases	Working automation with reproducible checks and safe failure behavior
Communicate a defensible finding	Select evidence → create a clear view → answer challenge questions	Accurate conclusion that survives an unfamiliar follow-up question

Appendix B Example Evidence Summary

Field	Illustrative value
Skill	Detect and repair inconsistent units in tabular data
Current state	Demonstrated; confidence moderate
Strongest evidence	Unassisted cleaning and SQL summary on an unfamiliar inventory dataset, rubric 4 of 5
Supporting evidence	Explanation correctly described type, unit, and validation decisions
Contradictory evidence	Needed two hints on an earlier duplicate-record task
Coverage gap	No evidence yet for multi-table joins with missing keys
Next action	Transfer task using clinic-supplies records
Review date	In 14 days to test retention
Limit	This claim concerns a narrow data-cleaning skill, not general data-analysis proficiency

Appendix C Definition of Done for Pilot MVP

- A learner can complete the diagnostic, receive a transparent skill map, and begin a recommended activity.
- The learner can complete the core loop in airplane mode using a downloaded content pack.
- The prototype is deployed through Firebase or Cloud Run and demonstrates a meaningful, inspectable Gemini or Gemma capability.
- After reconnecting, locally committed progress synchronizes without losing or duplicating accepted evidence.
- Learning and Proof Mode events record permitted assistance and evidence provenance.
- At least three evidence modalities update the learner model through versioned rules.
- Every loop transition is reproducible from its evidence and policy version, and the learner receives a plain-language reason for the next activity.
- Low-confidence and disputed scores reach a reviewer queue with full context.
- The team can reproduce every mastery state from immutable evidence events and policy versions.
- Power interruption, application termination, and repeated synchronization do not lose or duplicate accepted evidence.
- Representative personal devices can be provisioned through downloaded or physically transferred content packs and complete the pilot without a school hub.
- The experience meets defined performance, storage, battery, and accessibility targets on supported low-cost hardware.
- The pilot includes a secure external transfer outcome and a pre-specified analysis plan.
- Privacy, accessibility, and model-risk reviews are completed before external recruitment.